

Lecture 1: A Little Introduction to Signals

CSE 219: Signals and Linear Systems

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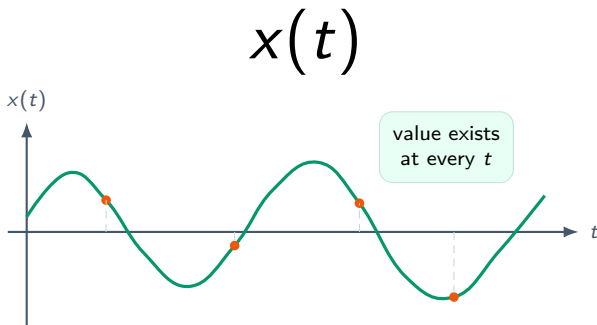
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**A signal is a pattern of change
that carries information**

Variation across time (voice, voltage) or space (image brightness)

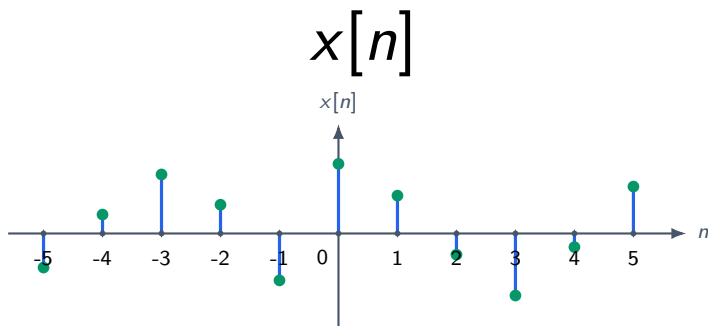
Continuous signals are smooth

- ▶ Signals that are **smooth**, defined at every instant
- ▶ Examples: microphone voltage, car velocity, temperature over time



Discrete signals have gaps between values

- ▶ Signals that are defined only at separated instants
- ▶ Digital audio **samples**, **daily** temperature log, **weekly** stock index



A little convention tells us which time points are allowed

Continuous signal

$$x(t)$$

Here, t can be any real value.

✓ $x(1)$ ✓ $x(2)$

✓ $x(0.5)$ ✓ $x(\pi)$

Discrete signal

$$x[n]$$

Here, n is an integer index.

✓ $x[1]$ ✓ $x[2]$

✗ $x[0.5]$ ✗ $x[\pi]$

Energy is simply a sum of squares

To measure how “large” a signal is, we square its values and add them up.

Continuous signal

$$E_{[t_1, t_2]} = \int_{t_1}^{t_2} |x(t)|^2 dt$$

Discrete signal

$$E_{[n_1, n_2]} = \sum_{n=n_1}^{n_2} |x[n]|^2$$

The absolute value lets the formula work for both real and complex signals.

Power is energy averaged over the interval

Continuous signal

$$P_{[t_1, t_2]} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} |x(t)|^2 dt$$

Divide the energy by the length of the time interval.

Discrete signal

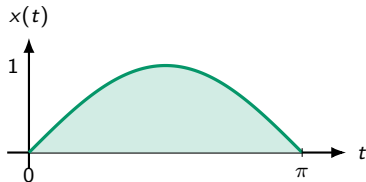
$$P_{[n_1, n_2]} = \frac{1}{n_2 - n_1 + 1} \sum_{n=n_1}^{n_2} |x[n]|^2$$

Divide the energy by the number of samples.

Continuous example: one half-sine has energy $\pi/2$

Signal

$$x(t) = \begin{cases} \sin t, & 0 \leq t \leq \pi, \\ 0, & \text{otherwise.} \end{cases}$$



Energy

$$\begin{aligned} E &= \int_0^{\pi} |\sin t|^2 dt \\ &= \int_0^{\pi} \sin^2 t dt \\ &= \frac{\pi}{2}. \end{aligned}$$

The signal is nonzero only from 0 to π , so we only integrate there.

Average power: let's divide by the interval length

Energy, we already found

$$E = \frac{\pi}{2}$$

Total squared size over $0 \leq t \leq \pi$.

Average power

$$P_{[0,\pi]} = \frac{E}{\pi - 0} = \frac{\pi/2}{\pi} = \frac{1}{2}.$$

Average squared size
over this interval.

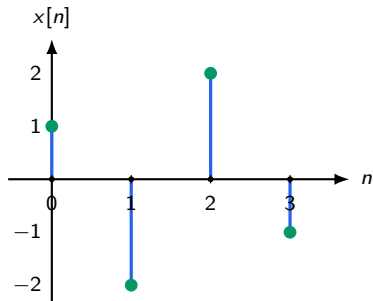
Discrete example: we start from the samples

Suppose a discrete-time signal has four samples:

$$x[0] = 1, \quad x[1] = -2, \quad x[2] = 2, \quad x[3] = -1.$$

Table of values

n	0	1	2	3
$x[n]$	1	-2	2	-1



Discrete example: square, add, then average

Energy

$$\begin{aligned} E &= \sum_{n=0}^3 |x[n]|^2 \\ &= |1|^2 + |-2|^2 + |2|^2 + |-1|^2 \\ &= 1 + 4 + 4 + 1 \\ &= 10. \end{aligned}$$

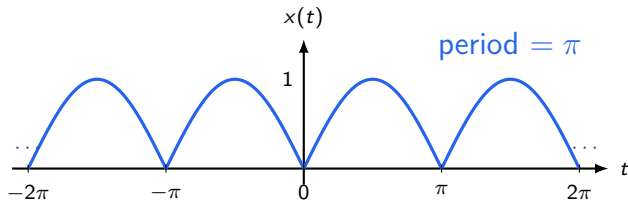
Average power

$$\begin{aligned} P_{[0,3]} &= \frac{1}{4} \sum_{n=0}^3 |x[n]|^2 \\ &= \frac{10}{4} \\ &= 2.5. \end{aligned}$$

Question: what happens if the half-sine repeats forever?

Previously, one half-sine over $0 \leq t \leq \pi$ had energy $\frac{\pi}{2}$.
Now suppose the same half-sine keeps repeating.

Periodic half-sine signal



Answer: infinite energy, but finite average power

Total energy

$$E = \frac{\pi}{2} + \frac{\pi}{2} + \frac{\pi}{2} + \dots$$
$$= \infty.$$

Each period contributes
the same positive energy.

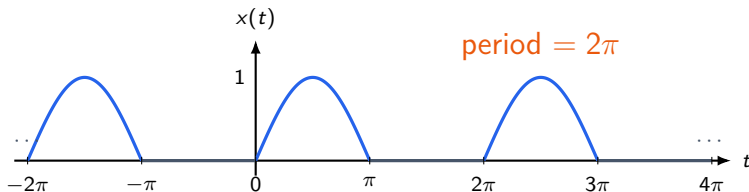
Average power

$$P = \frac{1}{\pi} \int_0^{\pi} \sin^2 t \, dt$$
$$= \frac{1}{\pi} \cdot \frac{\pi}{2}$$
$$= \frac{1}{2}.$$

Average **over one period** is enough.

Question: what if the half-sine rests for half the period?

Same lobe, but period 2π



Answer:

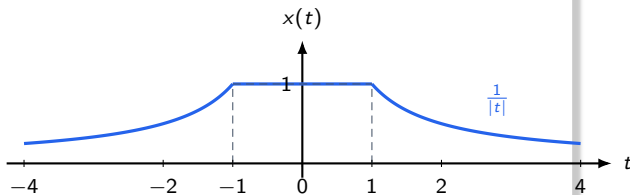
$$E = \infty, \quad P = \frac{1}{2\pi} \int_0^\pi \sin^2 t \, dt = \frac{1}{4}.$$

Can an infinite-range signal have finite energy?

Signal

$$x(t) = \begin{cases} 1, & |t| \leq 1, \\ \frac{1}{|t|}, & |t| > 1. \end{cases}$$

Graph of $x(t)$



Thanks to Dibyo and Mumtahina (CSE 23) for proposing this problem!

This infinite-range signal has energy 4

Energy

$$\begin{aligned} E &= \int_{-\infty}^{\infty} |x(t)|^2 dt \\ &= \int_{-1}^1 1^2 dt + 2 \int_1^{\infty} \left(\frac{1}{t}\right)^2 dt \\ &= 2 + 2 \left[-\frac{1}{t} \right]_1^{\infty} \\ &= 2 + 2 \\ &= 4. \end{aligned}$$

Average power

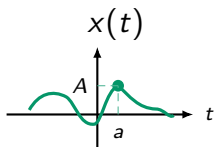
$$P = 0$$

Finite total energy spread
over infinite time gives
zero average power.

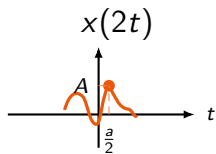
Transforming Signals

Signals feel nicer once we are able to move, flip, or stretch them as we wish.

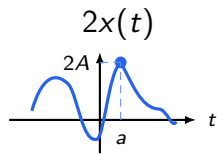
A signal can change both horizontally and/or vertically



value A occurs at $t = a$



Same value A
arrives at $t = \frac{a}{2}$



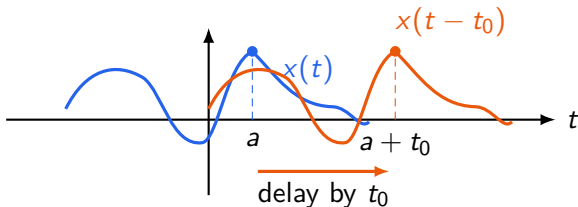
At same time $t = a$;
value becomes $2A$

$x(2t)$: time is compressed, so $a \mapsto a/2$.
 $2x(t)$: amplitude is doubled, so $A \mapsto 2A$.

Time shift: the whole signal moves left/right

When $t_0 > 0$

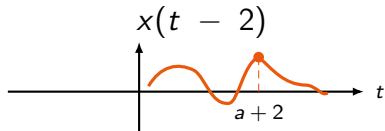
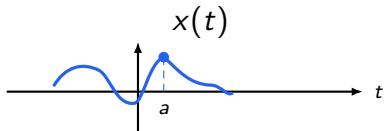
$x(t - t_0)$ is a **delay** to the right, while $x(t + t_0)$ is an **advance** to the left.



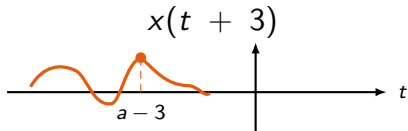
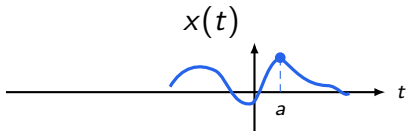
Same height, same shape but every value simply occurs t_0 later.

Shifting shifting: when right and when left?

Delay (shift right): $x(t) \longrightarrow x(t - 2)$

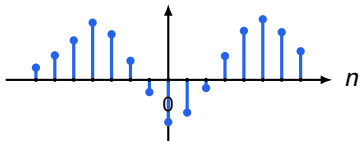


Advance (shift left): $x(t) \longrightarrow x(t + 3)$



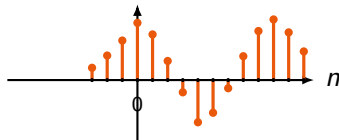
Shifting discrete signals: every value moves to a new index

Original: $x[n]$



→
delay by
4 samples

Delayed: $x[n - 4]$



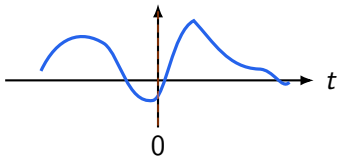
Time reversal: mirroring a signal with respect to the origin



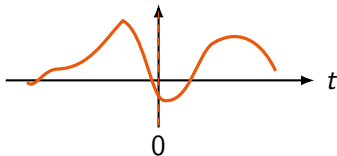
Time reversal: mirroring a signal with respect to the origin

Replace t by $-t$. A feature at $t = a$ appears at $t = -a$.

Original signal: $x(t)$

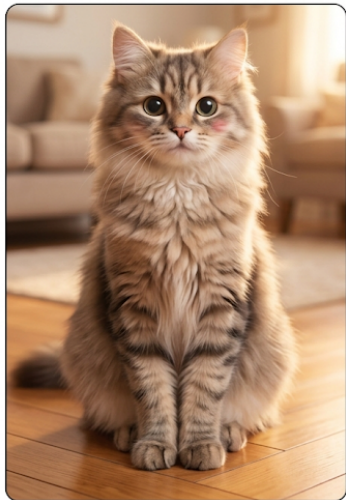


Reversed signal: $x(-t)$



Time scaling: compress or stretch?

$x(t)$



$x(2t)$



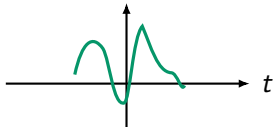
$x(t/2)$



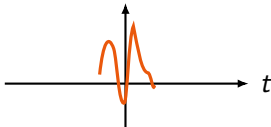
Time scaling: compress or stretch?

Time scaling changes along the horizontal axis, not the signal height.

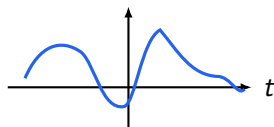
Original: $x(t)$



Compressed: $x(2t)$



Stretched: $x(\frac{t}{2})$



Can we combine transformations? Yes, sure!

Goal: obtain $x(2t + 3)$

$$x(t) \xrightarrow{\text{shift left by 3}} x(t+3) \xrightarrow{\text{compress along time by 2}} x(2t+3)$$

First create $x(t + 3)$. Then replace its time variable t by $2t$.

The second operation acts on the entire intermediate signal $x(t + 3)$.

Changing the order gives a different answer

$$x(t) \xrightarrow{\text{shift left by 3}} x(t+3) \xrightarrow{\text{compress time by 2}} \boxed{x(2t+3)}$$

Reverse order

$$x(t) \xrightarrow{\text{compress time by 2}} x(2t) \xrightarrow{\text{shift left by 3}} \boxed{x(2(t+3))}$$

The shift is now also multiplied by 2, so the result is different.

A simple recipe to obtain
 $x(\alpha t + \beta)$ from $x(t)$

Step 1

Apply the left/right shift part first:

$$g_1(t) = x(t + \beta)$$

Step 2

Apply time reversal only when $\alpha < 0$:

$$g_2(t) = \begin{cases} g_1(t), & \alpha > 0, \\ g_1(-t), & \alpha < 0. \end{cases}$$

Step 3

Finally, scale time by $|\alpha|$:

$$g_3(t) = g_2(|\alpha|t) = x(\alpha t + \beta)$$

A simple recipe to obtain $x(\alpha t + \beta)$ from $x(t)$

$$x(t) \longrightarrow x(t + \beta) \longrightarrow x(\alpha t + \beta)$$

$\beta > 0 \Rightarrow$ shift left in 1st step

$\beta < 0 \Rightarrow$ shift right in 1st step

$\alpha > 1 \Rightarrow$ time compression in 2nd step

$0 < \alpha < 1 \Rightarrow$ time stretching in 2nd step

$\alpha < 0 \Rightarrow$ **need to do time reversal before scaling**

Please write the intermediate signal before applying the next operation.

Example: obtain $x(-3t + 2)$ from $x(t)$

Step 1

Shift left by 2:

$$g_1(t) = x(t + 2)$$

Step 2

Apply time reversal (since $-3 < 0$):

$$g_2(t) = g_1(-t) = x(-t + 2)$$

Step 3

Compress time by 3:

$$g_3(t) = g_2(3t) = x(-3t + 2)$$

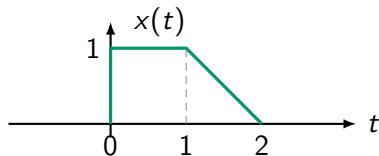
Visual example: begin with a piecewise signal

We will transform one simple piecewise signal step by step.

Original signal

$$x(t) = \begin{cases} 1, & 0 \leq t < 1, \\ 2 - t, & 1 \leq t \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Graph of $x(t)$



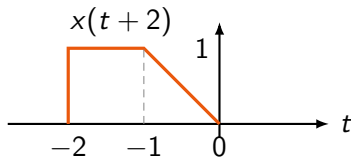
Step 1: shifting left by 2 gives $x(t + 2)$

Replace t by $t + 2$. Every feature moves 2 units to the left.

Shifted signal

$$x(t+2) = \begin{cases} 1, & -2 \leq t < -1, \\ -t, & -1 \leq t \leq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Graph of $x(t + 2)$



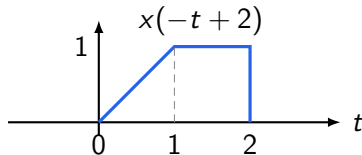
Step 2: reversing time gives $x(-t + 2)$

Now replace t by $-t$ in the shifted signal $x(t + 2)$.

Reversed signal

$$x(-t+2) = \begin{cases} t, & 0 \leq t < 1, \\ 1, & 1 \leq t \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Graph of $x(-t + 2)$



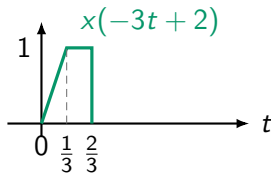
Step 3: compressing by 3 gives $x(-3t + 2)$

Finally, replace t by $3t$. The reversed signal becomes three times narrower.

Final signal

$$x(-3t+2) = \begin{cases} 3t, & 0 \leq t < \frac{1}{3}, \\ 1, & \frac{1}{3} \leq t \leq \frac{2}{3}, \\ 0, & \text{otherwise.} \end{cases}$$

Graph of $x(-3t + 2)$



Practice 1

Starting from $x(t)$, how can we obtain

$$y(t) = x\left(3\left(2 - \frac{t}{2}\right)\right)?$$

Hint: simplify the argument first, then work from the inside outward.

Practice 1: solution

$$\text{First simplify: } x\left(3\left(2 - \frac{t}{2}\right)\right) = x\left(6 - \frac{3t}{2}\right)$$

- ▶ Shift left by 6:

$$x(t) \longrightarrow x(t + 6)$$

- ▶ Reverse time:

$$x(t + 6) \longrightarrow x(-t + 6)$$

- ▶ Compress time by $\frac{3}{2}$:

$$x(-t + 6) \longrightarrow x\left(-\frac{3}{2}t + 6\right)$$

Practice 2

Start with

$$y_0(t) = x(t)$$

Apply the following operations in the given order and obtain the final signal.

1. Stretch time by a factor of 2.
2. Shift left by 3.
3. Compress time by a factor of 4.
4. Reverse time.

Practice 2: solution

- ▶ Stretch by 2:

$$y_1(t) = x\left(\frac{t}{2}\right)$$

- ▶ Shift left by 3:

$$y_2(t) = y_1(t + 3) = x\left(\frac{t + 3}{2}\right)$$

- ▶ Compress by 4:

$$y_3(t) = y_2(4t) = x\left(\frac{4t + 3}{2}\right) = x\left(2t + \frac{3}{2}\right)$$

- ▶ Reverse time:

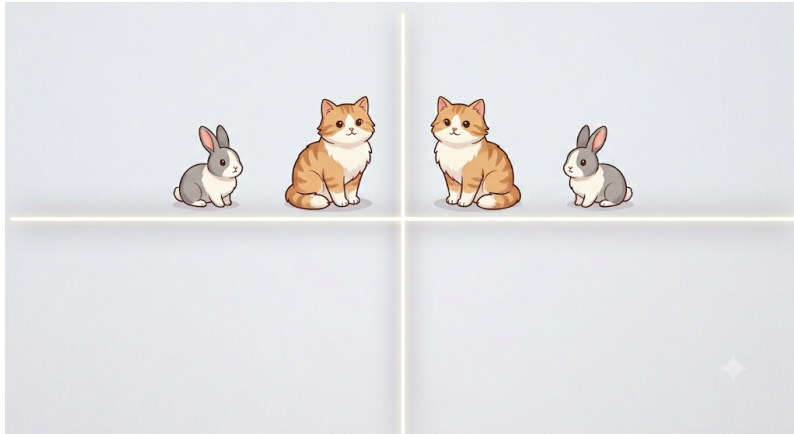
$$y_4(t) = y_3(-t) = x\left(-2t + \frac{3}{2}\right)$$

Even and Odd Signals

Even signals are mirror-symmetric about the vertical axis

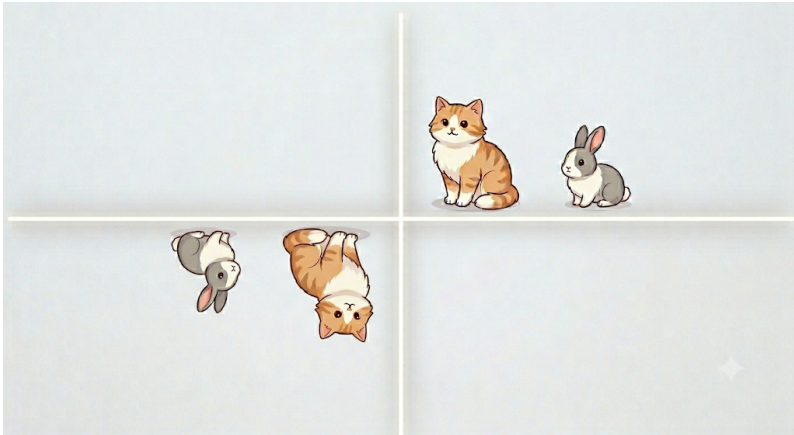
Property: $x(-t) = x(t)$,

Example: $x(t) = \cos(t)$



Odd signals flip sign across the origin

Property: $x(-t) = -x(t)$, **Example:** $x(t) = \sin(t)$



Every signal has an even part and an odd part. Really!

Even part

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)]$$

Odd part

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)]$$

Quick check: does it really work?

Check 1

Show that the even part is truly even:

$$x_e(-t) = x_e(t)$$

Check 2

Show that the odd part is truly odd:

$$x_o(-t) = -x_o(t)$$

Check 3

Show that, $x(t) = x_e(t) + x_o(t)$

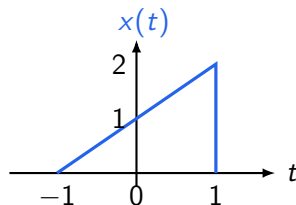
Example: begin with a signal that is neither even nor odd

Original signal

$$x(t) = \begin{cases} t + 1, & -1 \leq t \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

This signal is neither even nor odd.

Graph of $x(t)$



We first reflect the signal to obtain $x(-t)$

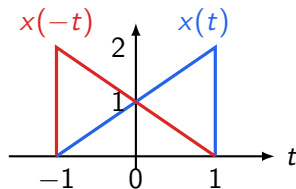
Time-reversed signal

For $-1 \leq t \leq 1$,

$$x(-t) = 1 - t$$

The graph of $x(-t)$ is obtained by reflecting $x(t)$ about $t = 0$.

$x(t)$ and $x(-t)$

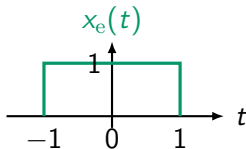


Adding and averaging extracts the even part

Add, then average

$$\begin{aligned}x_e(t) &= \frac{1}{2}[x(t) + x(-t)] \\&= \frac{1}{2}[(t+1) + (1-t)] \\&= 1\end{aligned}$$

Even part: $x_e(t)$

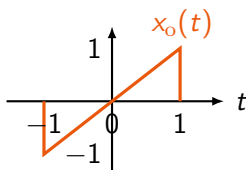


Subtracting and averaging extracts the odd part

Subtract, then average

$$\begin{aligned}x_o(t) &= \frac{1}{2}[x(t) - x(-t)] \\&= \frac{1}{2}[(t+1) - (1-t)] \\&= t\end{aligned}$$

Odd part: $x_o(t)$

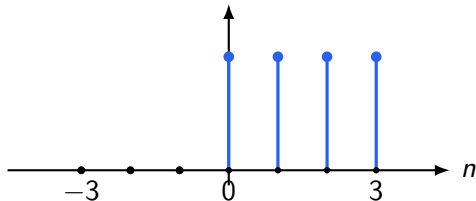


Another Example: Discrete-Time Signal

Original signal $x[n]$

$$x[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

Graph of $x[n]$

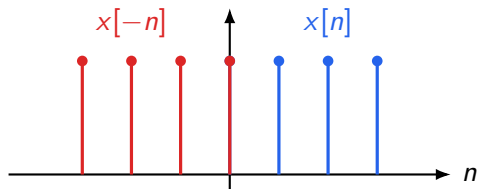


Step 1: Time Reversal $x[-n]$

Time-reversed signal

$$x[-n] = \begin{cases} 1, & n \leq 0 \\ 0, & n > 0 \end{cases}$$

$x[n]$ and $x[-n]$



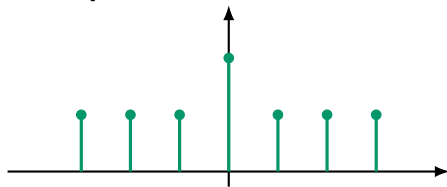
Step 2: Even Component

Even part

$$x_e[n] = \frac{1}{2}(x[n] + x[-n])$$

$$x_e[n] = \begin{cases} \frac{1}{2}, & n \neq 0 \\ 1, & n = 0 \end{cases}$$

Even part



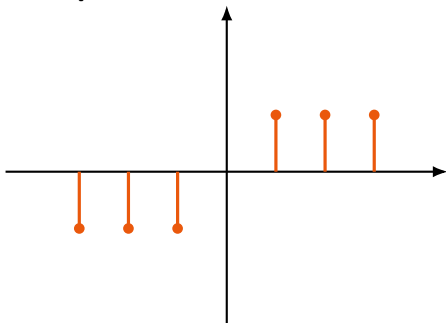
Step 3: Odd Component

Odd part

$$x_o[n] = \frac{1}{2}(x[n] - x[-n])$$

$$x_o[n] = \begin{cases} -\frac{1}{2}, & n < 0 \\ 0, & n = 0 \\ \frac{1}{2}, & n > 0 \end{cases}$$

Odd part



Thank you!